Result 9.4

$$\lim_{x\to 0}\frac{\log(1+x)}{x}=1.$$

Proof

$$\text{Take} \quad \log(1+x) = y$$

Then $y \to 0$ as $x \to 0$ and

$$1+x = e^y$$

$$x = e^y - 1$$

$$\text{Therefore,} \quad \lim_{x\to 0}\frac{\log(1+x)}{x} = \lim_{y\to 0}\frac{y}{e^y-1}$$

$$= \lim_{y\to 0}\frac{1}{\left(\frac{e^y-1}{y}\right)}=\frac{1}{1}=1. \qquad \square$$

Some important limits without proof

Results 9.5 to 9.9

(5) $\lim\limits_{x\to 0}\dfrac{\sin^{-1}x}{x}=1$

(6) $\lim\limits_{x\to 0}\dfrac{\tan^{-1}x}{x}=1$

(7) $\lim\limits_{x\to\infty}\left(1+\dfrac{1}{x}\right)^x$ exists and this limit is e.

(8) $\lim\limits_{x\to 0}(1+x)^{1/x}=e$

(9) $\lim\limits_{x\to\infty}\left(1+\dfrac{k}{x}\right)^x=e^k.$

This number e is also known as transcendental number in the sense that e never satisfies a polynomial (algebraic) equation of the form

$$a_0x^n+a_1x^{n-1}+\cdots+a_{n-1}x+a_n=0.$$

Example 9.32

Evaluate : $\lim\limits_{x\to 0}(1+\sin x)^{2\operatorname{cosec} x}$

Solution

Let $\sin x = \dfrac{1}{y}$

As $x \to 0$, $y \to \infty$ and

$$\lim_{x\to 0}(1+\sin x)^{2\operatorname{cosec} x} = \lim_{y\to\infty}\left(1+\frac{1}{y}\right)^{2y} = \left[\lim_{y\to\infty}\left(1+\frac{1}{y}\right)^{y}\right]^2 = e^2 .$$

Example 9.33

Evaluate : $\lim\limits_{x\to\infty}\left(\dfrac{x+2}{x-2}\right)^x$.

Solution

$$\lim_{x\to\infty}\left(\frac{x+2}{x-2}\right)^x = \lim_{x\to\infty}\left(\frac{x-2+4}{x-2}\right)^{x-2+2} = \lim_{x\to\infty}\left(1+\frac{4}{x-2}\right)^{(x-2)+2}$$

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